

Little Caesars

How the reduction is calculated

4 October 2026. Numbers below are the home card for closed day 3 October 2026. Times are Asia/Riyadh.

1. What Reduction means

Reduction is the percent change when this period used less electricity than the period it is compared with. Increase is the same formula when this period used more. The minus sign stays on the percent.

change kWh = this period - previous period

change % = (this period - previous period) / previous period x 100

The percent is rounded to a whole number. The kWh figures are shown to one decimal. If there is no previous period, the card does not show a percent. If the previous value is 0 and this value is 0, the change is 0%. If the previous value is 0 and this value is not 0, the change is 100%.

2. One closed day

An energy day runs from 06:00 to 06:00 the next morning. On the afternoon of 4 October, 4 October is still open, so the latest closed day is 3 October: 06:00 on 3 October through 06:00 on 4 October.

device kWh = meter at the next 06:00 - meter at this 06:00

day total = Lights + AC + Oven + Freezer + Chiller

The five tiles are that one day: Lights 50.1, AC 82.4, Oven 147.3, Freezer 32.5, Chiller 21.4. Added as shown, those round to 333.7 kWh. The card shows 333.8 kWh because the day total is rounded on its own, separately from each tile. Those tiles are not used in the reduction percents.

A missing next reading stays blank. It is not treated as zero. The result is never negative.

3. Last 7 days: Reduction -15%

This is the seven closed days ending on 3 October: 27 September through 3 October. The total is 2,440.2 kWh.

The previous week is the seven days immediately before that: 20 September through 26 September. That total is 2,873.0 kWh.

2,440.2 - 2,873.0 = -432.8 kWh

-432.8 / 2,873.0 x 100 = -15%

The card shows Reduction -15%, then 2,440.2 kWh, then -432.8 kWh vs previous week. This is a rolling seven days. It is not the Saturday-to-Friday week.

4. This month: Reduction -62%

This month is only the closed days so far: 1, 2, and 3 October. The total is 1,103.0 kWh. 4 October is still open, so it is not included.

Last month, on this card, is the same day numbers in September: 1, 2, and 3 September. It is not the whole of September. That total is 2,938.6 kWh.

$$1,103.0 - 2,938.6 = -1,835.6 \text{ kWh}$$

$$-1,835.6 / 2,938.6 \times 100 = -62\%$$

The card shows Reduction -62%, then 1,103.0 kWh, then -1,835.6 kWh vs last month.

5. Savings in SAR

The home card does not show money. The chain board does. Inside one month, the first 6,000 kWh cost 0.22 SAR each. Every kWh after that costs 0.32 SAR. Exactly 6,000 kWh is still entirely at 0.22.

$$\text{cost} = \min(\text{kWh}, 6000) \times 0.22 + \max(0, \text{kWh} - 6000) \times 0.32$$

Saved kWh compares this month so far with the same days last month and the same days last year, then takes the average of those two gaps.

$$\text{saved kWh} = \text{average of (last month same days - this month) and (last year same days - this month)}$$

$$\text{saved SAR} = \text{cost}(\text{this month} + \text{saved kWh}) - \text{cost}(\text{this month})$$

A positive saved kWh means this month is below that average. The SAR figure is the bill difference for that gap, using the tariff above.

6. Saved by automation

This is the green line on the hourly chart, separate from the home-card reduction. It estimates kWh an AC did not use after its target was raised, or after it switched to fan only. Off is not counted.

For each AC, the occupied target is the lowest target seen while the mode was cool or heat.

A 15-minute slot saves when the mode is fan only, or when the mode is cool or heat and the target is more than 0.4 C above that occupied target. Example: occupied target 20 C, empty-room target 26 C.

The comparison rate is how fast that AC was cooling at the occupied target during the previous 2 hours. 0.25 hours is one 15-minute slot (15 / 60).

$$\text{cooling rate} = \text{average of (kWh used in the slot} / 0.25) \text{ over those normal slots}$$

$$\text{saved kWh in the slot} = \max(0, \text{cooling rate} \times 0.25 - \text{kWh used})$$

The green point for an hour is the sum of those slot savings in that hour. Nothing is counted until a normal cooling rate exists.